

Process, structure & task challenge

Structure and process

	Day 1 (Tue)	Day 2 (Wed)	Day 3 (Thurs)	Day 4 (Fri)		Day 5 (Mon)	Day 6 (Tue)	Day 7 (Wed)
8:00 a.m. - 12:00 p.m.	Morning Introduction Speech	Morning Module Interaction	Morning Module Localization	Morning Module Mechane Perception	WEEKEND	Morning E-Test Speech Introduction Challenge	Morning Challenge Processing	Morning Challenge Processing
1:00 p.m. - 5:00 p.m.	Afternoon Module Kinematics	Afternoon Module Control	Afternoon Module Navigation	Afternoon Speech		Afternoon Challenge Processing	Afternoon Challenge Processing	Afternoon Presentation Results Challenge

The goal of the challenge is to enable you to apply the knowledge and skills you have learnt over the past few days in a practical environment. The task of the therefore developed challenge is described in detail below. You will work on the challenge as a group. Each group will have its own playing field and its own robot. To complete the eight tasks, you have two full days. The tasks can be completed independently and in any order. However, it is recommended to proceed in chronological order, as the level of difficulty increases with each task. If you have any questions about the individual tasks, the internship supervisors will be happy to help you.

On Wednesday afternoon, the individual groups will present their results. The aim is for your robot to fulfil the tasks as quickly as possible. You have three attempts for each task. If you are unable to perform a task, you may move on to the next task and restart the process at this point. Points are awarded for successfully completed tasks. Bonus points are awarded based on the total time to complete all tasks. If you must touch your robot during the tasks, you will receive penalty points for each interaction. The team with the most points earns fame and honour!

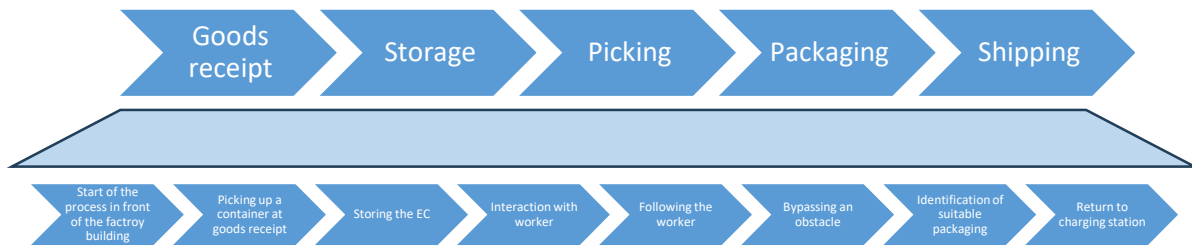
Task

You are a team of consultants from the company "RoboTUM". Recently you have received a new order. You are to automate the entire internal logistics process at Meyer & Meyer. An initial internal workshop has shown that the use of mobile robots offers great potential.

Colleagues of yours have already presented a concept based on this to the customer during a meeting. The customer was enthusiastic about the basic idea, but doubted whether it could be implemented in their own company.

Your task is therefore to recreate the exemplary logistics process and thus demonstrate that the use of robots would be possible at Meyer & Meyer. Your management thought it would be a good idea to divide you and your colleagues into groups. The best concept will be presented to the customer at the end.

The overall process at the customer can be seen in the following diagram:



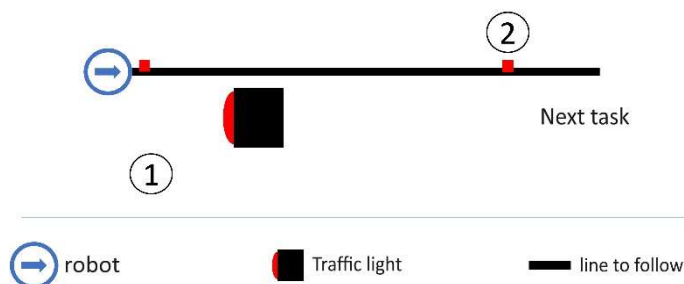
On the following pages you will find a detailed description of the tasks, including illustrations of each subtask. As mentioned above, the level of difficulty increases with each additional task. When programming the tasks, use the functions and code blocks you have learnt in the modules.

General information

- The robot drives along the black lanes on the entire factory premises, unless explicitly stated otherwise.
- The red markers with a number next to them indicate the start of a new task. The robot should read the number of the task from a QR code located on the left side of the lane. The Challenge.py template serves as a distribution function and calls the respective task.

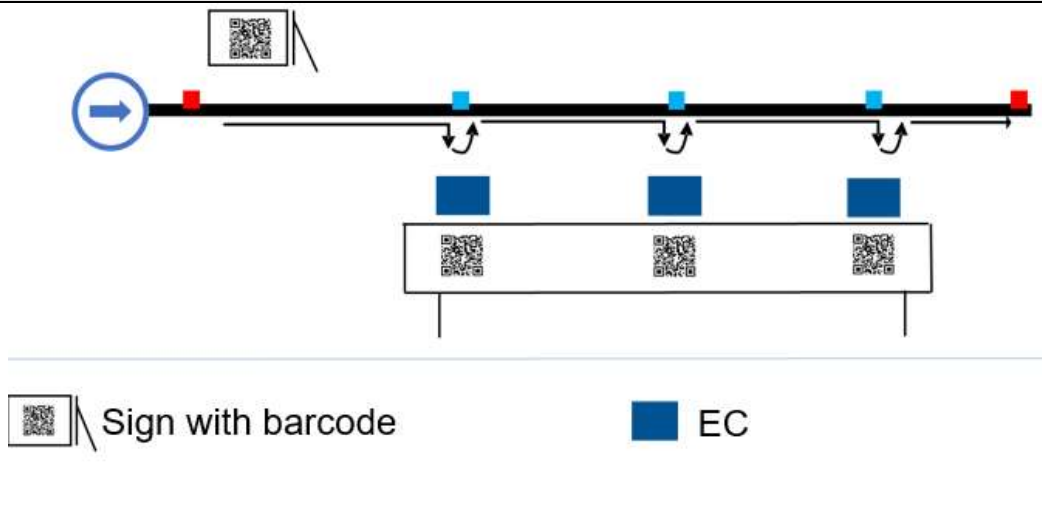
Task 1 – Starting the process in front of the factory building

At Meyer & Meyer, the process begins in front of the factory building. The tugger train, which in the future is to be replaced by an automated guided vehicle (AGV), must wait in front of the factory gate until the traffic light turns green because of crossing traffic. Then it can continue its journey.



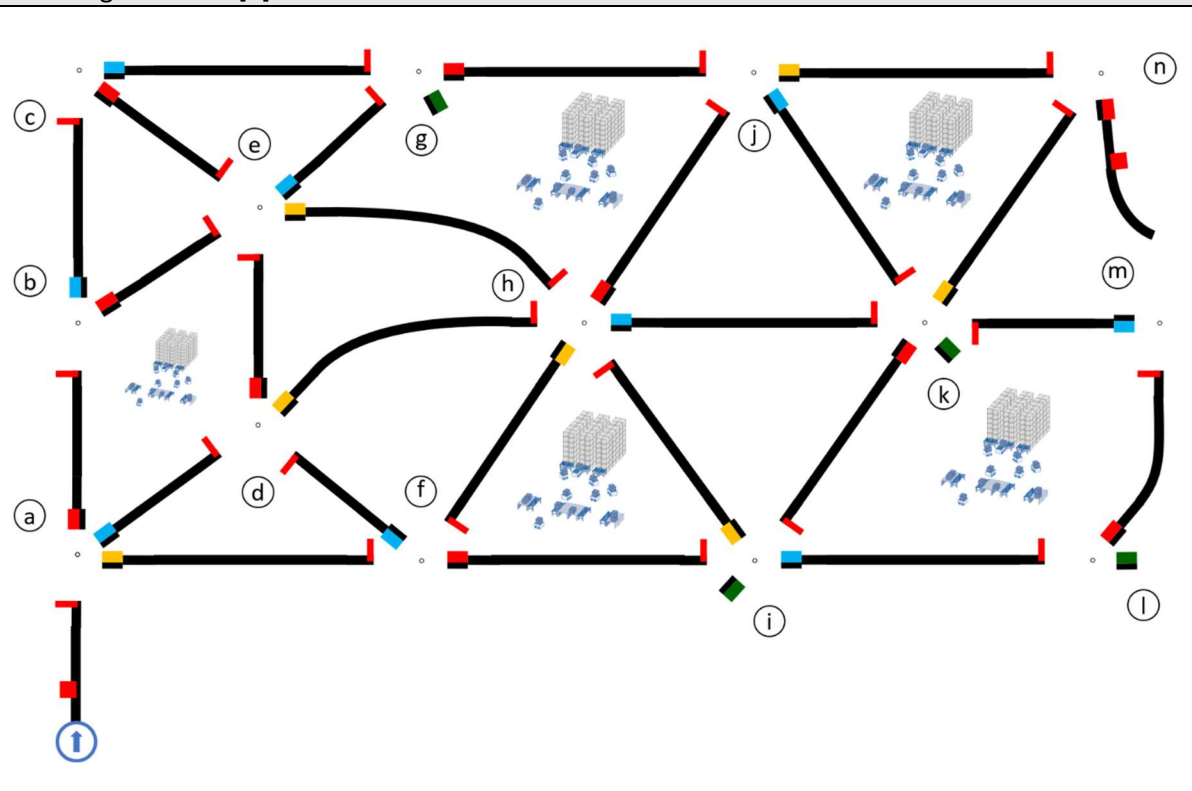
Task 2 – Picking up a container at goods receipt

The tugger train then picks up a container with raw components at the goods receipt. In the future, the container will be identified using a barcode that is presented to the robot when it enters the goods receipt. The same barcode is also located on the correct container. By comparing both barcodes, it will be possible to prevent picking up the wrong container.



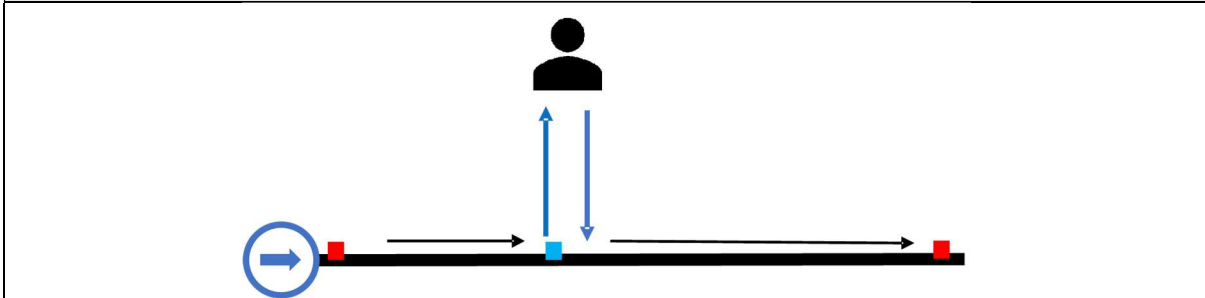
Task 3 - Storing the EC

After the AGV has picked up the EC, it must be stored for later use. The storage location to be approached must be specified in the Python code before the program starts. The storage locations [g, i, k, l], which are marked by a green marker, are available for selection. The robot can only move along the black lines in the storage area. The lines represent the aisles of the warehouse. As soon as the EC has been unloaded at the correct storage location, the robot should leave the warehouse at storage location [n].



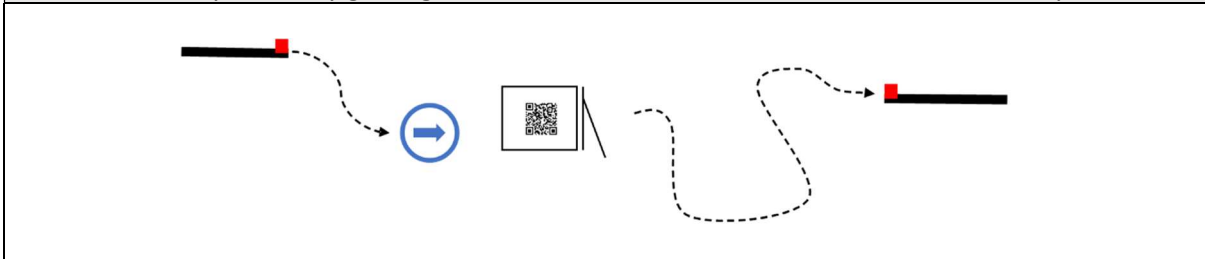
Task 4 – Interaction with worker

The fourth process step consists of the AGV providing support for order picking. To do this, it uses MQTT to send an arrival message at the blue marker and waits for a worker. The worker hands the robot an empty container and determines the correct packaging shape for shipping. This information is sent to the robot via MQTT.



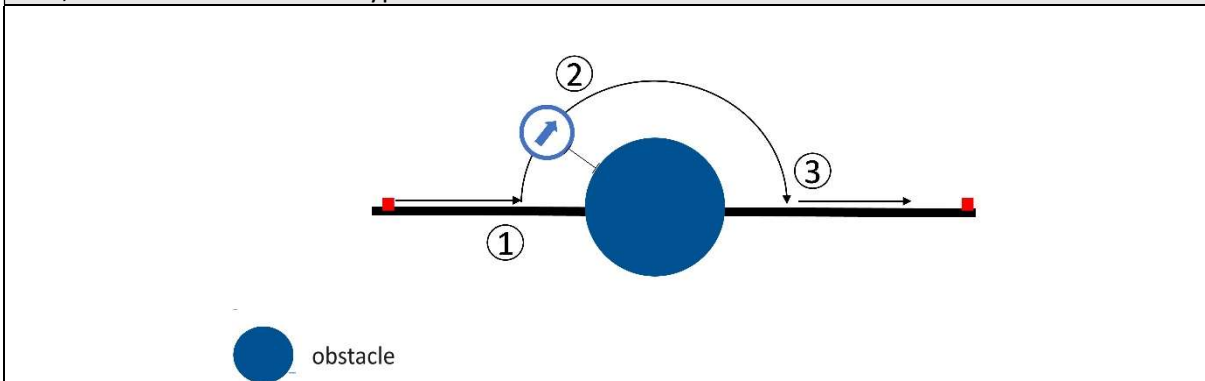
Task 5 – Following the worker

Inside the picking area, the AGV follows the worker as he picks the required items and places them in the container. The worker is recognised by a QR code. The technical feasibility of this task is to be demonstrated by manually guiding the QR code so that the robot follows it to the final position.



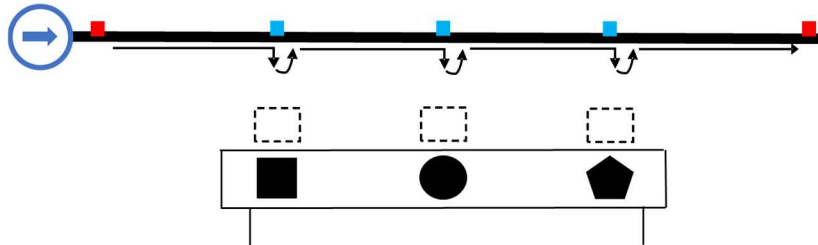
Task 6 – Bypass an obstacle

The robot now independently brings the items picked by the worker to the company's shipping area. To do this, it must cross an open area where there may be obstacles in the way. If this happens, the AGV should first pause for 10 seconds to give the obstacle (person, pallet, etc.) the opportunity to leave the route again. If this does not happen, the bypassing process should be initiated. To do this, the obstacle should be bypassed with a maximum distance of 30 cm.



Task 7 – Identification of suitable packaging

In the company's shipping area, the robot should find the correct packing station for the picked items. To do this, it uses the previously transmitted packaging shape, which can be used to differentiate between the stations. Once the robot has found the correct station, it places the EC there and a worker takes over the packaging process.



Task 8 – Return to the charging station

Once the robot has delivered the picked items to the shipping area, it drives to its charging station. The ground on the way there is often very dirty starting at the blue marking, which is why navigation is carried out without lane markings from this point onwards. Obstacles can also block the AGV on this route. In this case, it should wait until the obstacle has disappeared and only then continue its journey.

